Roll No. | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 13

B.Pharma (Sem.–1)

PHARMACEUTICAL ANALYSIS-I

Subject Code : BP-102T

M.Code : 74645

Date of Examination: 12-01-2023

Time : 3 Hrs. **Max. Marks : 75**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. Explain briefly :

(a) Define as amphiprotic solvent. Give its example.

(b) What is the pH of a 0.01 N solution of NaOH?

(c) What is a mixed indicator? Give an example.

(d) Write the primary standard and indicator used in diazotization titration.

(e) Differentiate between iodometery and iodimetery.

(f) What is co-precipitation and post-precipitation?

(g) What is peptisation in gravimetry?

(h) Differentiate between accuracy and precision.

(i) You are provided a 35% w/v solution of HC1. Calculate the volume of this solution to prepare a 500 ml of N/20 solution of HCI.

(j) Define error.

SECTION-B

2. What is the principle of potentiometric titrations? Explain the construction, working, merits and demerits of various reference electrodes used in this method.

3. Classify complexometric titrations. Give examples in each category.

4. (a) '*Comment upon various solvents used in non-aqueous titrations.*' Give examples.

   (b) Plot the neutralization curve for the titration of a strong acid with a weak base.

SECTION-C

5. Enumerate various steps involved in estimation of Barium sulphate via gravimetry.

6. Explain in detail the different theories of acid base indicators.

7. Write a note on primary and secondary standard. Give examples.

8. How do you prepare a 0.1 M solution of Ceric ammonium sulfate? Explain its standardization giving balanced chemical equations and general calculation.

9. Comment upon different types of errors and their sources.

10. Give a detailed account on assay of sodium benzoate.

11. What is the principle of polarography? Explain their applications.

12. Give the construction and working of calomel electrode with the help of neat diagram.

13. Comment upon the concept of oxidation and reduction.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.